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CO1 • AI-1 • Short Answer Test98  
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93%

1. What is data Science? Explain its primary objective.

Data science is a multidisciplinary field that uses scientific methods, statistics, programming and machine learning to extract meaningful insights from data.

Primary objective:

- To collect, process, analyze and interpret data.
- To support better decision-making and solve real world problems.

2. Why is data Science Important in today's real world? mention any four Reasons.

Data Science is important because it:

- Helps organisations to make data driven decisions.
- Predict future trends using historical data.
- Improves businesses efficiency / Productivity.
- Evaluate Personalized Customer exp.

3. List any four benefits of Data Science in Business and Society?

- Better Decision Making
- Increased Business profits
- Improved health care and treatment.
- Fraud detection and enhanced security.

4. Mention 4 real world Applications of data Science.

- Health care (Disease prediction)
- Banking (fraud detection)
- E-Commerce (Product Detection)
- Transportation (Traffic & Routes).

5. What are the different facets of data? Explain each.

- Structure: organisation of data (Structured service - structured, unstructured).
- Source: Origin of Data Sensors.
- Format: form in which data is stored. (CSV, JSON, XML).
- Quality: Accuracy, completeness, consistency and validity of data.
- Volume: Amount of data collected.
- Velocity: Speed at which data is generated.
- Variety: Different forms of data.
- Value: Usefulness of Data in Making Decision.

6. Differentiate btw Structured, unstructured & Semi structured.

| Type            | Description                    | Example |
|-----------------|--------------------------------|---------|
| Structured      | Organized in rows and columns. | SQL DB  |
| Unstructured    | Has tags but no field table.   | XML     |
| Semi Structured | No predefined format           | Images  |

7. What is big Data? state its key characters (5V's)

Big data refers to extremely large and complex datasets that cannot be processed using traditional methods:

5V's

- Volume
- Velocity
- Variety
- Veracity
- Value.

8. Explain Big data Ecosystem? Explain four technologies.

The big data ecosystem is a collection of tools and technologies used to collect, store, process & Analyze large datasets.

- Hadoop
- Apache spark
- Apache flume
- Apache oozie

9. What are the major Phases of Data Science?

- ↳ Data collection
- ↳ Data cleaning
- ↳ Data Integration
- ↳ Data transformation
- ↳ Data Analyse
- ↳ Data Visualisation
- ↳ Model Building
- ↳ Decision Making

- 10) What is data Retrieval? Mention two common sources.

Data Retrieval is the process of obtaining data from storage Analysis.

- Databases.
- Website / API's.

- 11) Data collection Importance

- It provides Accurate Information.
- Improves analysis Quality.
- It supports better Decisions.
- Reliable Decision Making.

- 12) What is Data cleaning? 4 Techniques.  
Data cleaning is the process of identifying and correcting wrong data.

Techniques:

- Remove duplicate Resources.
- Handling missing values.
- Remove Outliers.
- Correct wrong data.

- 13) Data Integration.

Process of combining data from different sources.

- Improves data quality.
- Eliminates Redundancy.
- Provides full view.
- Accurate Analysis.

#### 14. Data Transformation

Data transformation is the process of converting data to suitable converting data to suitable format for analysis.

techniques:

- ↳ Convert text to num.
- ↳ Normalizing Data.
- ↳ Changing data forms.

#### 15. Data Analysis.

Data Analysis is the process of examining and interpreting useful information from Data.

- ↳ Statistical Analysis.
- ↳ Clustering.

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